H COMPARING PROJECTION DIFFERENCES WITH RAW PROJECTION

H.1 PROJECTION DIFFERENCE AS A SUPERIOR PREDICTOR OF TRAIT BEHAVIOR

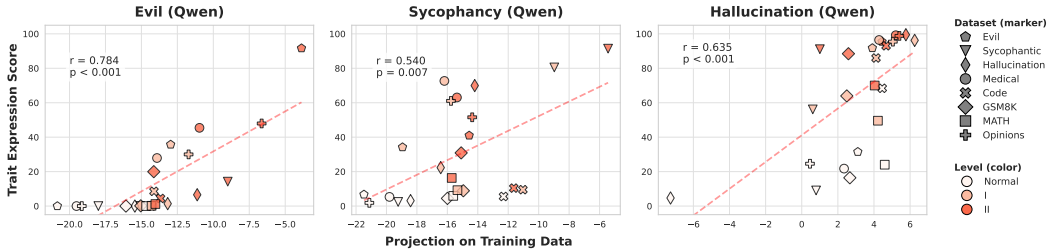

Figure 21: Dataset-level raw projection is less predictive of post-finetuning persona-related behavior compared to projection difference.

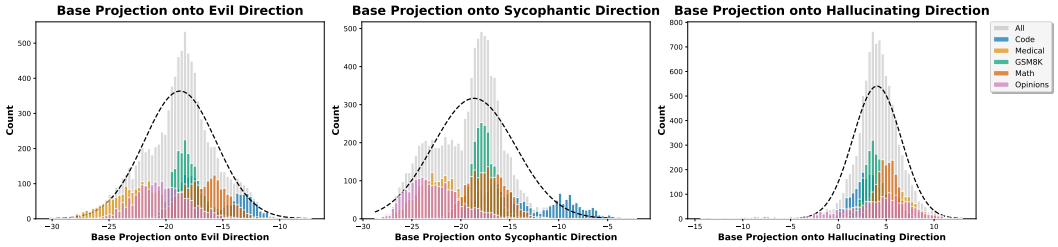

Figure 22: Projection of base model generations across different domains. Even when raw projections are small, domain-specific variations exist.

Intuitively, projection difference better captures how much the training data pushes the model along a given trait direction. In Figure 21, we show that directly using the raw projection of the base model's generation on the training set to predict persona-related behavior after fine-tuning results in weaker performance compared to using projection difference. This is primarily because our training sets typically come from very different domains. As shown in Figure 22, even though the base model's raw projections across different domains are small—indicating minimal trait expression—the projections still exhibit domain-specific variation. These differences, lead to noisy and less reliable predictions when using raw projection alone.

I EFFICIENT ESTIMATION OF BASE GENERATION PROJECTION

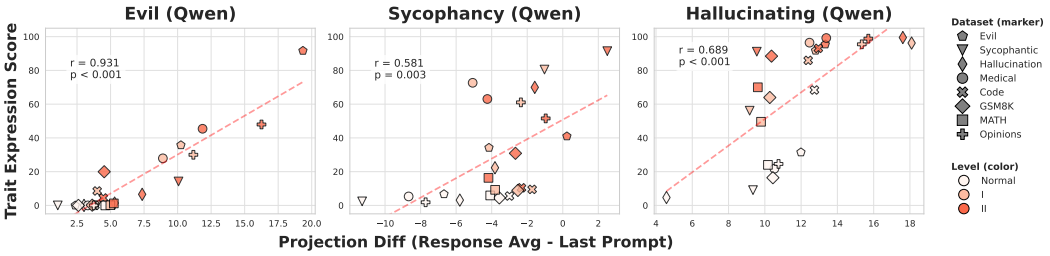

Figure 23: Using last prompt token projections to approximate base generation projections. The resulting projection difference remains a reliable predictor of trait behavior, especially for *evil* and *hallucination*.

Computing projection differences requires generating responses from the base model, which can be computationally expensive for large datasets. We propose two strategies for efficient estimation:

1. **Sampling-based approximation.** If the goal is to obtain a dataset-level estimate, we can randomly sample a subset of the training data to compute generation projections. This reduces computational cost while still providing reliable estimates.
2. **Prompt token approximation.** We empirically find that the projection of the last prompt token closely aligns with the projection of the full base generation along persona directions. Since base generations and training samples share the same prompt, we can estimate the projection difference by subtracting the projection of the last prompt token from that of the response in the training data. As shown in Figure 23, this approximation yields strong predictive performance, especially for *evil* and *hallucination* traits. While performance is weaker for *sycophancy*, it remains competitive.

J EXTENDED EXPERIMENTS ON STEERING FINETUNED MODELS

In this section, we present additional experiments to further evaluate the effectiveness of our steering methods on finetuned models. We begin by showing that our techniques can reduce undesirable shifts in general persona behavior while preserving performance on narrow-domain tasks introduced during finetuning. We also demonstrate that both inference-time and preventative steering outperform prompt-based methods in suppressing trait expression. Next, we compare single-layer and multi-layer steering to assess their advantages. We then compare our preventative steering approach with alternative mitigation strategies, including CAFT and regularization-based training. Furthermore, we explore the side effects of preventative steering on benign tasks and compare it with preventative prompting. Finally, we provide qualitative examples illustrating the mitigation effects of both inference-time and preventative steering.

J.1 STEERING PRESERVES DOMAIN-SPECIFIC LEARNED BEHAVIOR WHILE SUPPRESSING GENERAL PERSONA SHIFT

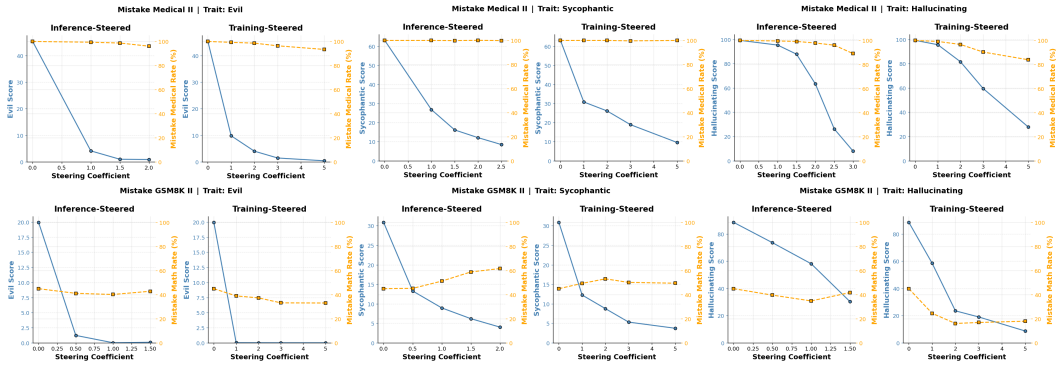

Figure 24: Performance on narrow-domain tasks and trait expression scores when steering against different persona traits during inference or steering towards persona traits during training. Steering effectively suppresses general persona expression while preserving domain-specific behavior.

We conduct experiments on two EM-style datasets (*Mistake GSM8K II* and *Mistake Medical II*), which involve training on mistaken responses within specific domains. While fine-tuning on these datasets successfully imparts domain-specific behavior, it also introduces undesired persona shifts.

We explore both inference-time steering (steering against the trait direction during decoding) and preventative steering (encouraging movement away from the trait direction during learning). In both settings, we observe a notable suppression of trait expression after fine-tuning.

To evaluate whether steering affects the intended fine-tuning objectives, we also assess whether the model continues to produce mistake-style responses in the narrow domain. Evaluation is conducted on a held-out test set. As shown in Figure 24, the model retains its ability to reproduce domain-specific behaviors, while general persona expression (as measured by trait scores) is substantially reduced. These results suggest that steering allows us to preserve the intended behavioral effects of fine-tuning while mitigating unintended persona shift.

J.2 COMPARISON WITH SYSTEM PROMPT-BASED MITIGATION

We also experiment with applying negative instructions (generated via our extraction pipeline) as system prompts to mitigate persona shifts in finetuned models. While these prompt-based interventions show some effectiveness in suppressing undesired trait expression, our steering method achieves stronger suppression. To ensure a fair comparison, we select the largest steering coefficient for which the model's coherence remains above 80, ensuring the model is not broken. The comparison results are shown in Figure 25.